Part A – Learning Objective 3.2: I can derive the array factor for an arbitrary linear array and apply pattern multiplication.

Show your work. A **Documentation** line at the end is required (write **None** if you did not collaborate).

1. **Reading a sweep.** A lab group sweeps the PHASER with all eight elements active and a source at boresight. Working at 10.3 GHz with element spacing $d = 14$ mm, they record the trace below. The sweep step is 2.8125° and the peak reads -4.1 dBFS.

θ (°)	0.0	2.8	5.6	8.4	11.3	14.1	16.9	19.7	22.5	25.3
Power (dBFS)	-4.1	-4.6	-6.2	-9.1	-14.3	-24.1	-22.0	-17.4	-16.6	-18.1

   (a) Using linear interpolation between the two samples that bracket the half-power level, find the half-power beamwidth. The trace is symmetric about 0°.

   θ_{HP} = ____

   (b) Find the first-null beamwidth as the sweep reports it, and compare it with FNBW $= 2 \arcsin(\lambda / N d)$.

   $\text{FNBW}_{\text{meas}}$ = ____

   $\text{FNBW}_{\text{calc}}$ = ____

   (c) Find the first sidelobe level relative to the peak, in dBc.

   SLL = ____

   (d) No sample in the sweep reads below about -24 dBFS, which is 20 dB below the peak, yet the noise floor on this kit is 23 dB below the peak. Explain in one or two sentences why the deepest sample sits 3 dB above the floor rather than on it.

2. **Predict before you measure.** The same array is reconfigured by setting some element gains to zero. The spacing $d = 14$ mm and the frequency 10.3 GHz do not change, so $\lambda = 29.1$ mm throughout.

   (a) Rx1, Rx2, Rx7 and Rx8 are set to zero, leaving Rx3 through Rx6 active — four adjacent elements. Predict the half-power beamwidth from $\theta_{\text{HP}} \approx 0.886\lambda/(Nd)$.

   θ_{HP} = ____

   (b) Now only Rx3, Rx4 and Rx5 are left on — three adjacent elements. Predict the half-power beamwidth and the first-null beamwidth.

   θ_{HP} = ____

   FNBW = ____

   (c) By how many dB should the peak of the three-element sweep sit below the peak of the full eight-element sweep, if every element gain that is still on remains at 100%?

   Δ = ____

   (d) A student proposes reaching the same 35° beamwidth by leaving all eight elements on and turning every gain down to 37.5%. Explain in one or two sentences why that does not work.

3. **Reconciling a sidelobe.** A second group measures the first sidelobe of the full eight-element array at -11.5 dBc, while the array factor for $N = 8$ predicts -12.8 dB. Their noise floor sits at -23 dBc and the sweep step is $2.8125°$.

   (a) The floor adds noise power to the sidelobe power the receiver reports. Treating the two as uncorrelated, compute the level the sweep should report for a true -12.8 dB sidelobe sitting on a -23 dBc floor.

   SLL = ____

   (b) Their peak reading is also affected. The peak is 23 dB above the floor by definition. By how much does the floor lift the peak reading, and which way does that push the reported sidelobe level in dBc?

   peak lift = ____

   (c) The floor explains 0.4 dB of the 1.3 dB discrepancy. Name two other contributions that plausibly make up the rest, and say for each whether it would move the reading up or down.

4. **The two-element case.** With only Rx4 and Rx5 active the array has two elements at $d = 14$ mm, and at 10.3 GHz the array factor is $AF = \cos(\psi/2)$ with $\psi = kd\sin(\theta)$.

   (a) Find the half-power beamwidth exactly, without using the $0.886\lambda/(Nd)$ rule.

   θ_{HP} =

   (b) Show that the $0.886\lambda/(Nd)$ rule fails here, and state the percentage error it makes.

   error =

   (c) Evaluate $\lambda/2d$ and use it to explain why the lab table lists the first-null beamwidth of this array as 180°.

   $\lambda/2d$ =

   (d) How far below its own peak does this two-element pattern fall at $\theta = 90^\circ$? Compare that with the 23 dB of dynamic range the sweep has above the noise floor at eight elements, and say what the trace will look like near the edges of the scan.

   drop at 90° =

5. **Design judgment.** Your section is asked to demonstrate a 20° beam on the PHASER at 10.3 GHz, using only the Element Gains controls.

   (a) How many adjacent elements must be active to give a half-power beamwidth closest to 20°? Give the aperture and the resulting beamwidth.

   N = ____

   θ_{HP} = ____

   (b) What does this cost in peak received level compared with the full eight-element array, and how much dynamic range is left above the noise floor?

   Δ = ____

   range = ____

   (c) A classmate suggests instead applying a Hann taper to all eight elements, since that also widens the beam to about 19.5°. State one advantage and one disadvantage of the taper against switching elements off, and say which you would choose for a demonstration whose purpose is to show that beamwidth depends on aperture.

Part B – Learning Objective 3.6: I can distinguish between array factor and true antenna pattern and account for element pattern effects.

6. **Reading dBc off a Tracking-tab trace.** A cadet walks an HB100 around the PHASER on a 1 m arc with the beam held at broadside and records the Signal vs Time trace. The three lobe peaks read -18.4 dBFS (main lobe), -31.2 dBFS (left first sidelobe), and -29.9 dBFS (right first sidelobe).

   (a) Convert both sidelobe readings to dBc.

   L_{left} =

   L_{right} =

   (b) The cadet's partner raises **Rx Gain** by 6 dB and repeats the walk. State what happens to the three dBFS readings and what happens to the two dBc values, and explain why the two answers differ.

   (c) A uniform 8-element array has its first sidelobes at -13 dBc. Comment on the pair measured above: is this array uniformly illuminated, and what accounts for the 1.3 dB spread between the two sides?

7. **Why the time axis is not an angle axis.** The same trace runs from $t = 0.00$ s, when the source was at -90°, to $t = 7.40$ s, when it reached $+90^\circ$. The main-lobe peak lands at $t = 3.10$ s. The beam was commanded to broadside.

   (a) Explain why the lobe *amplitudes* on this trace are usable while the lobe *angles* are not.

   (b) Assume a constant walking speed and convert the main-lobe time to an angle. Compare it against the commanded beam angle and state the error.

   θ = [ ]

   error = [ ]

   (c) The true half-power beamwidth is 13.1°. If the hand moves as much as 30% faster or slower than its average while crossing the main lobe, give the range of beamwidths a constant-speed conversion would report.

   range = [ ]

   (d) Name the piece of range hardware that removes this error and state what it records that the hand walk does not.

8. **Scan loss and sidelobe asymmetry.** The projected-aperture rule puts the peak power gain of a steered array at $\cos\theta_0$ relative to broadside.

   (a) Compute the predicted peak drop at steer angles of 30°, 45°, and 60°.

   30° =

   45° =

   60° =

   (b) A broadside walk peaks at -18.4 dBFS and a 30° walk peaks at -19.3 dBFS. Give the measured scan loss, and state whether one pair of walks resolves it given ± 0.5 dB of run-to-run scatter.

   Δ =

   (c) The measured 30° trace shows first sidelobes of -12.2 dBc on the low-angle side and -15.2 dBc on the high-angle side. Explain why they are unequal, given that the array factor alone would make them equal.

   (d) Real patch elements follow a power pattern nearer $\cos^{1.4}\theta$ than $\cos\theta$. Recompute the 60° scan loss and state how much worse it is than the ideal-element bound.

   loss =

9. **How far out is far enough.** The PHASER's 8 elements are spaced 14 mm apart, so the aperture spans $D = 98$ mm between the outer element centers. The HB100 runs at 10.525 GHz.

   (a) Find the far-field distance for this aperture and state whether the 1 m arc used in lab is far enough.

   r =

   (b) A Ku-band panel with a 0.200 m aperture is to be measured at 15.0 GHz. Find its far-field distance.

   r =

   (c) The Ku-band panel is measured on the same 1.0 m classroom arc anyway. Find the phase error across the aperture edge and compare it against the 22.5° tolerance behind the far-field criterion.

   $\Delta\phi$ =

   (d) Describe what a 90° edge phase error does to the measured pattern, and say which pattern feature it corrupts worst.

10. **Classroom against chamber.** The lab trace carries about ± 1 dB of ripple from room reflections adding to the direct path.

    (a) List which of the four hand-measurement artifacts from the lab an anechoic chamber removes, and which one it does not remove by itself.

    (b) A single reflected path of relative amplitude ρ adds to a direct path of amplitude 1. Find the ρ that produces a $+1$ dB peak in the received amplitude, and express it in dB.

    ρ = ____

    (c) The main-lobe peak and the sidelobe peak each carry their own ± 1 dB of ripple. Give the worst-case uncertainty on a sidelobe level reported in dBc, and give the range a true -13 dBc sidelobe could read.

    uncertainty = ____

    range = ____

    (d) A customer requires written proof that this array's sidelobes are below -25 dBc. State whether the classroom measurement can support that claim, and give the two reasons that decide it.

Documentation: _____